MSc in Marketing Analytics - EDHEC

Data Visualisation Project

4 consulting-style case studies: enriched, open SNCF data

Associated workbook: Donnees_SNCF_Projet_Dataviz_vf.xlsx

1. The dataset

SNCF is France's national railway company, operating most of the country's rail services, including TGV high-speed trains, and maintaining much of the rail network.

Main source: SNCF's open data portal (ressources.data.sncf.com), extracted on July 22, 2026 via the Explore v2.1 API. The data is raw and unprocessed: cleaning and joining the tables is part of the exercise.

Worksheet	Content	Granularity / Period
Regularite_TGV	Monthly punctuality by TGV route: scheduled trains, cancellations, departure/arrival delays, 6 delay causes.	Monthly × route - Jan. 2018 → March 2026 (130 directional lines)
Frequentation_gares	Annual station ridership, segmented by station type (Halte, Proximité, Ville, Hub, LGV, IDF) and regional division.	Annual × station - 2015 → 2024 (~3,000 stations)
Tarifs_TGV	Minimum/maximum price by route, carrier (TGV INOUI / OUIGO / OUIGO Classic Train), class and fare profile.	Route × class × profile (36,349 rows)
Emissions_CO2	Carbon footprint compared across train / plane / car / coach, with associated distances.	119 routes (TGV, TER, Intercités, International)
Liste_gares	Station reference table: UIC code, municipality, department, geographic coordinates.	~6,500 stations
NPS_Satisfaction_SIMULEE ⚠	Satisfaction score (NPS), dominant trip purpose, and complaint reasons, by directional route. SIMULATED data for teaching purposes.	130 directional lines, same grain as Regularite_TGV
Rentabilite_Couts_SIMULEE ⚠	Estimated margin by route (revenue - cost). Distance and price modelled from real data; capacity, load factor and staff/maintenance costs SIMULATED.	130 directional lines, same grain as Regularite_TGV

A point of caution: station identifiers are not consistent

Three join keys coexist in this workbook, with very different levels of reliability:

- Tarifs_TGV and Emissions_CO2 carry a UIC code - the join to Liste_gares works for ~96% of cases.
- Frequentation_gares carries a UIC code (code_uic_complet) - the join to Liste_gares works for ~99% of cases.
- Regularite_TGV (and therefore also NPS_Satisfaction_SIMULEE, which reuses its grain) carries no UIC code at all: only station names in uppercase without accents are available (e.g. "PARIS MONTPARNASSE"), whereas the other tabs use mixed-case labels with variable spelling ("Paris Montparnasse", "Paris-Gare-de-Lyon", "Paris-Nord" vs "Paris Nord"...).

Any question that joins Regularite_TGV with another tab therefore requires building a correspondance table (mapping) between station names - this is an expected methodological step, not a technical detail to work around.

Warning - the NPS_Satisfaction_SIMULEE table

⚠ Simulated data

• NPS_Satisfaction_SIMULEE contains data that is ENTIRELY SIMULATED for teaching purposes (NPS score, number of reviews, dominant trip purpose, complaint reasons). None of these values come from a real SNCF survey.

• The grain is identical to that of Regularite_TGV (gare_depart, gare_arrivee): 130 directional lines, corresponding exactly to the 65 TGV routes tracked by Regularite_TGV, in both directions. This is not a sample deliberately restricted in size: it is the entire universe of major TGV routes covered by this workbook.

• This table contains no recalculated performance column: to compare real performance with perceived satisfaction, it must be joined to Regularite_TGV on (gare_depart, gare_arrivee), and the performance indicator must be built from scratch. This is part of the expected work, not data to be read directly.

• Any deliverable using this table must explicitly restate its simulated nature.

Warning - the Rentabilite_Couts_SIMULEE table

⚠ Partially simulated data

• Rentabilite_Couts_SIMULEE estimates a margin by route (estimated revenue - estimated cost), using a mixed method that should be clearly distinguished:

• Real or calculated: distance (calculated by geolocating stations); average price (modelled from a regression actually calculated on Tarifs_TGV prices, ≈ €13.90 + €0.114/km); infrastructure toll (€9.50/km) and energy cost (€4.30/km), anchored on published, order-of-magnitude public figures (SNCF Réseau tolls approved by the ART regulator, published TGV electricity consumption).

• SIMULATED, with no public source available at this level of detail: average train capacity, load factor, and staff/maintenance cost.

• No revenue, cost, or margin column is provided: this is a calculation expected from teams, based on the input columns — revenue = trains × capacity × load factor × price; cost = trains × distance × (toll + energy + staff/maintenance cost); margin = revenue - cost.

• Any deliverable using this table must restate this distinction and discuss how sensitive the result is to the simulated assumptions, rather than presenting the margin as an established fact.

2. General instructions for teams

Organisation: Each team chooses one of the four proposed case studies.

Deliverables:

- **Tableau Packaged Workbook (.tbwx file)**, containing all the visual elements created for the study.
- **Report with:**
    - **Explanation of the rationales** behind all the decisions made to create the visual elements. The limitations of the analysis must be made explicit (join quality, sample size, missing data, bias, and, where relevant, the simulated nature of a dataset).
    - **Recommendation notes** addressed to the fictional "client" described in the brief.
- **Presentation to be used during the last session**: **10 minutes** per team maximum.

Method: The question is deliberately open-ended. It's up to each team to define the relevant angles of analysis, clean and combine the necessary data, and justify their choice of metrics. **There is no single right answer**.

3. The case studies

Case 1 - The TGV's value proposition

Context - SNCF Voyageurs' marketing and CSR department must decide between growing the low-cost OUIGO offering, its fare policy (including student fares), and highlighting the environmental argument against air travel, cars and carpooling. It wants an objective view of the TGV's current value proposition before deciding.

> ***What is the TGV's real value proposition compared with its alternatives, and how should SNCF evolve it?***

Case 2 - The rail network versus the territory

Context - SNCF Gares & Connexions is preparing its five-year station investment plan under budget pressure. It must decide on investment priorities at the national, territorial scale.

> ***How well does the rail network serve the French territory today, and where should SNCF prioritise its investment?***

Case 3 - The traveller satisfaction paradox

Context - SNCF Voyageurs' customer relations department has noticed a gap between the performance indicators it communicates and feedback from the field (complaints, reviews). Ahead of its next quality roadmap, it wants to understand this gap rather than keep steering solely on operational indicators.

> ***To what extent does the performance SNCF measures match what travellers actually experience, and what should that change about its priorities?***

Case 4 - The TGV network's real profitability

Context - SNCF Voyageurs' finance department is preparing a decision on whether to maintain, strengthen, or reorganise certain routes.

> ***Which TGV routes are profitable, which are not, and what does that imply for SNCF's priorities?***